Applied Machine Learning

Evaluation of Scorers and Rankers

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CHALMERS

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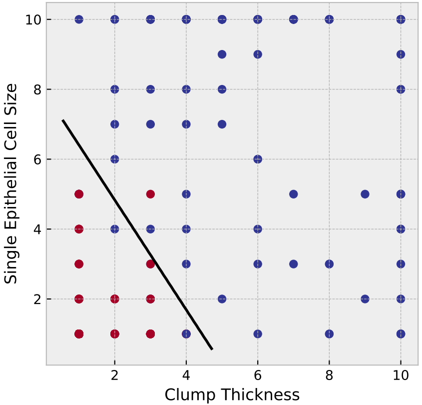
what if it's more important to find the malignant tumors?

- how can we adjust our decisions?

breast cancer example

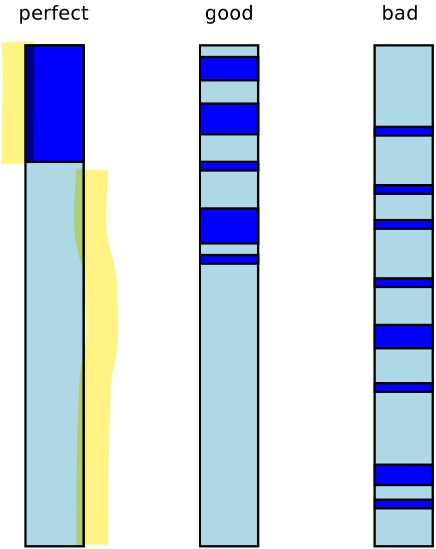


when we evaluate, can we abstract away from the threshold?

ranking and scoring systems

- many classifiers output some sort of **confidence score**
  - `decision_function`, `predict_proba`, ...
- the methods we discuss now are more or less the same as those used to evaluate **search engine** (or more generally, **ranking systems**)

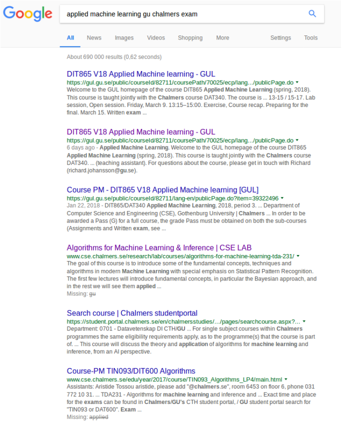
intuition: evaluating rankers



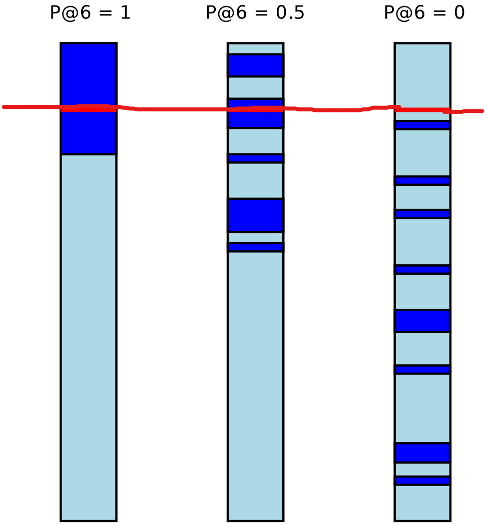
evaluating a search engine: quality of the first page

- when I use a search engine, how good are the results on the first page?
- **precision at k**

$$P@k = \frac{\text{nbr relevant items in top } k}{k}$$



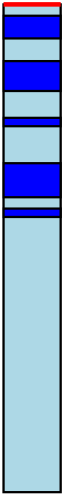
evaluating a search engine (another example)

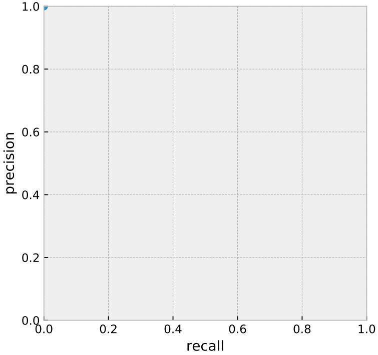


precision / recall curve

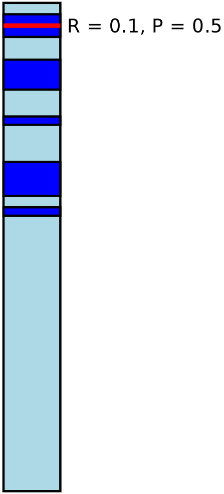
- is there a way to evaluate ranker or scorer that does not depend on the threshold?
- in a **precision/recall curve**, we compute the precision for different recall levels
  - that is, for different thresholds
- we visualize the relationship using a plot
  - good = curve is near top right

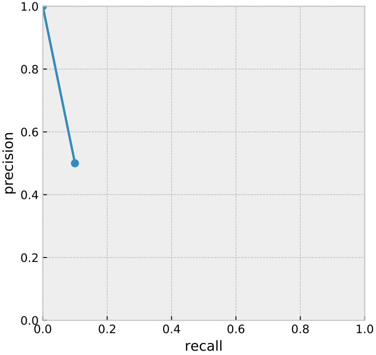
P / R curve (example)



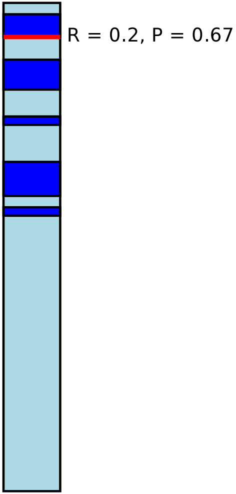


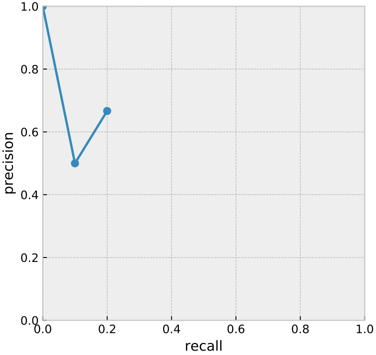
P / R curve (example)



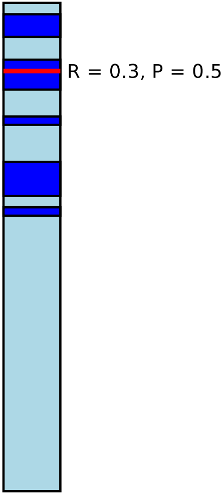


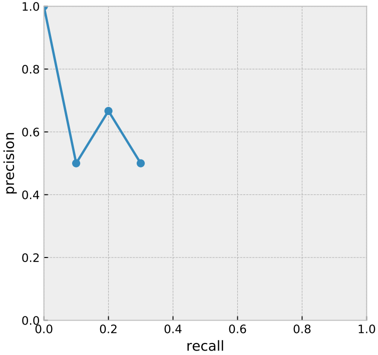
P / R curve (example)



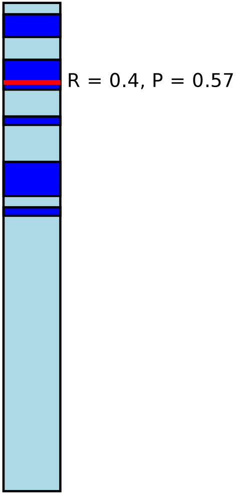


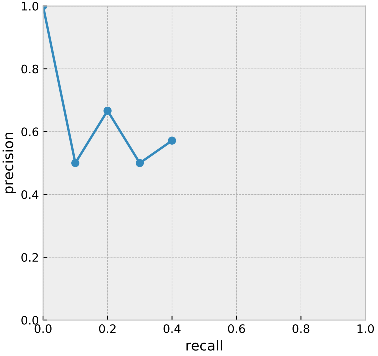
P / R curve (example)



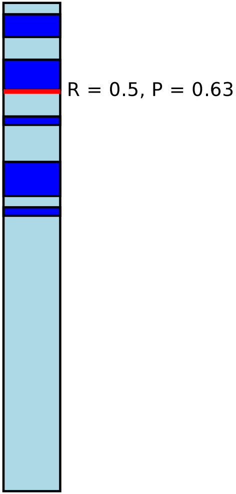


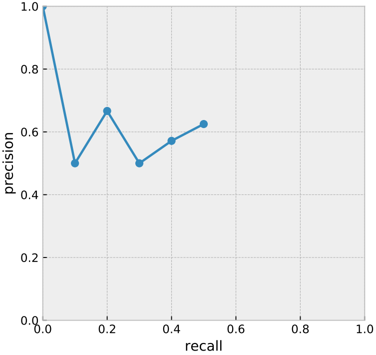
P / R curve (example)



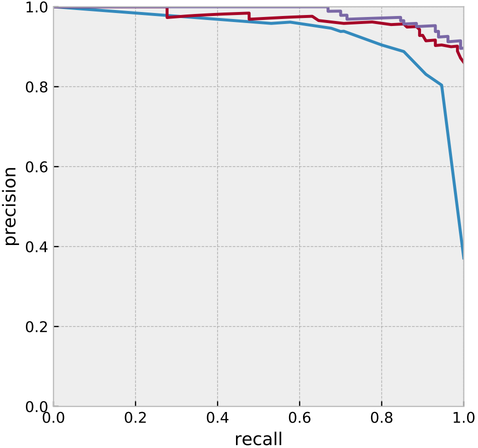


P / R curve (example)

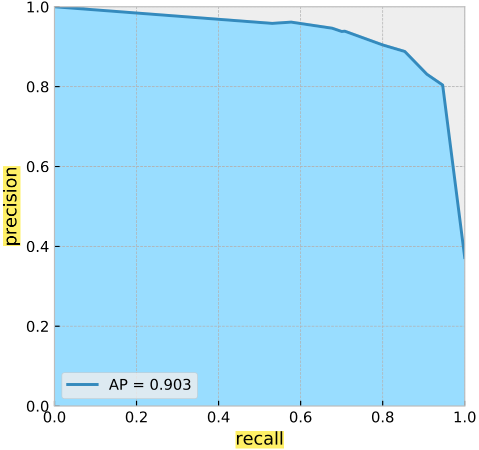




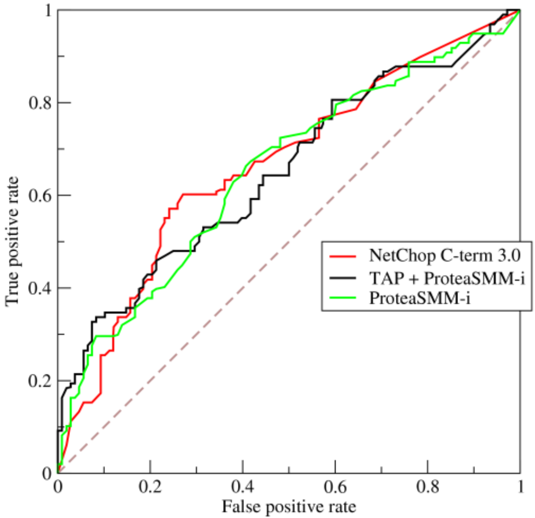
comparing P/R curves



average precision



ROC curve (Receiver Operating Characteristic)



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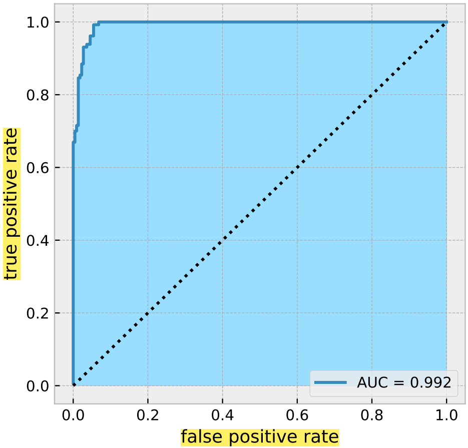
Area Under Curve score



summary: evaluating scoring and ranking systems

- regular classification metrics typically depend on a **threshold**
- we may want to consider the system's behavior for different values of the threshold
  - drawing a **curve** to explore the tradeoff
  - computing an **aggregate value** over all thresholds (**AP**, **AUC**)
- methods are similar to those used to evaluate **search engines**